Does the daily review teach better when it teaches the move a lesson teaches?

The lesson pick, its prompt and a check on verdict talk, replayed on twelve past days

Chess Coach research lab

Abstract

The app's daily routine writes a review of the player's games every day, with a narrated video. He asked it to stop telling him what the engine thinks of a position and to teach the way people learn: at each key moment, a sound move a player at his level can find, what it does on the board, and the idea behind it from the study lessons he follows. One change does all three parts: the review's `teach` command picks the move a lesson teaches among the moves the engine says hold for him, the routine's prompt asks for that move, what it does and the lesson's rule, and the routine's checker refuses a video line that says who is better. We replayed the whole review on twelve past days, once with today's code and once with the change, with the routine's own model, tools and checker. Every review passed the check. The change's videos spoke no verdict line on any day, against three to six a day today. Judging blind in both orders, a model preferred the change's coaching and its whole script on all twelve days, and both controls held. An audit against the evidence found fewer false claims in the change's scripts (0.42 per script against 1.17), with more misleading ones (3.17 against 2.25), mostly lesson rules and findability worded loosely. Its moment scenes said what the position is about before the move more often (70% against 53%), but what the better move does no more often (55% against 56%), so the share of the three moves he asked for rose by +7 (−5 to +17) points, an interval that does not clear zero. That is the one condition of the rule fixed before the first run that did not hold, so by its rule the change is not kept. A second round, designed after reading the first round's scene audits and run on the same days, added one sentence to the prompt: the scene that shows his move ends on the better move and what it does. With it every moment scene said what the better move does, the share of the asked moves rose by +17 (+6 to +27) points, the judge preferred the change on eleven of the twelve days, and every condition of the rule held, so the second round is kept. Its scripts carried more false claims than today's (1.50 per script against 1.17), inside the noise. The owner, watching a day replayed with it, found that it still names a lesson and plays a line without saying why each move works; the app's next version, which reads every move of both lines on the board, is not measured here.

1 The question

Each day the routine has a model review the day's games and write the annotations the app shows and a narrated video script: scenes with a board from one of his games and beats with a spoken line, a caption, the moves played on the board and pauses where he is meant to think. Before the change, the better move it taught at a turning point was the move players at his band most often choose among those the engine says hold, else the engine's move, and its videos often said who stood better: "level", "you are winning", "the engine agrees".

The player asked for the review to teach from his study lessons instead: a sound move he can find, what it does on the board and the idea behind it, from a lesson when one fits, even when it is not

the engine's first choice, as long as the engine says it keeps his position. The lesson-moves study measured the move such a review would teach on every turning point of his archive: a lesson teaches it at most turning points, it gives up a fraction of a win% against the engine's first line, and players at his band choose it more often than that first line. This study asks whether the review the routine writes with the change teaches better, without becoming less true or much longer.

2 The change

The change is the app at `f7a8b12`, in three parts that ship together:

- **The pick.** `coach review teach` reads what each move that holds for him is for (it meets a threat, wins material, trades queens, breaks with a pawn, puts a rook on an open file, and so on) and the lesson chapters that teach it, and teaches the move his band plays most often among those a lesson teaches for a purpose his move lacked, with the chapters and their rules.
- **The prompt.** The daily prompt asks for that move, what it does in teach's words and the chapter's rule in the writer's own, a moment that opens with what the position is about, and no line that says who is better.
- **The check.** `daily check` refuses a video line that says who is better, by the same rule this study counts verdict lines with. It applies from the day after the change ships; the replayed days are older, so the change's cases run an eval-only commit (`4d0e9d3`) that moves that date before them and changes nothing else.

Today's code is the app's main at `c98822d`.

3 The cases and the writer

A case is a past day whose review landed on the app's main with a video script: twelve days from October 13 to October 28, with one to 14 games each (62 in all). Each config has its own case of each day: a checkout of the app at the config's code with the coach data as main held it just before that day's review landed and the day's archived engine analysis, with the day's brief written again by that code's `coach daily prepare`, offline. The review itself, its annotations and its video script, is what a writer run writes. The split was fixed before any run, interleaved by date: train is Oct 13, Oct 15, Oct 18, Oct 20, Oct 24 and Oct 27; held out is Oct 14, Oct 17, Oct 19, Oct 22, Oct 25 and Oct 28.

The writer runs as the routine does: Opus 5.5, headless, with nothing inherited from the session that launched it, the study lessons' server attached as the live routine has it, and a short preamble that names the day, says which steps ran and asks for the review, done when the routine's check passes. Each config wrote every day once (24 runs in all).

4 The graders

A script with no model reads each run: the final check, the check rounds, files changed, commands it should not have run, how often it ran `teach`, the video lines that say who is better by the change's own rule (run on both configs' scripts), the annotations' moments that link a lesson, the script's shape, its length heard once (spoken words at 2.6 a second plus every pause), cost and minutes.

Sonnet 5.5 audits each video script alone against the evidence it was written from: the brief without its bulk, the day's games and turning points, the engine digest, how often players at his band find the engine's move, and for every board scene the position as the archive, the engine, Maia-2 and the board code read it, with the engine's best reply after each move the game did not play; and the study lessons, every lesson's title and what it teaches, with the sections of each lesson a script names. It lists the claims the evidence contradicts (*false*), does not carry (*unsupported*) or slants (*misleading*), and for each board scene its kind, its false board claims, whether it says who is better, and for a moment scene whether it says what the position is about before the move, what the better move does, an idea to reuse, a lesson and whether the lesson fits, whether the better move is sound, and the coaching moves of the earlier video study.

Sonnet 5.5 also judges each day's two scripts blind, in both orders, on truth, coaching, listening and overall, with the player's ask in its instructions; a side wins a criterion only when both orders agree. Two controls check the judge on every day: today's script against itself, and against a decoy that adds as many spoken words as the change added that day (at least 40), in beats that restate their scene with nothing new.

5 The rule

A day scores +1 when the change wins a criterion, −1 when it loses and 0 for a tie; intervals are 95% percentile bootstrap intervals over days. The change ships only when every condition of Table 1 holds, fixed before its first run.

Condition	Estimate (95% interval)	Holds
Coaching on the train days above 0	+1.00 (+1.00 to +1.00)	yes
Coaching on the held-out days above 0	+1.00 (+1.00 to +1.00)	yes
Overall not below 0	+1.00 (+1.00 to +1.00)	yes
False claims per script not up	−0.75 (−1.42 to −0.08)	yes
False board claims per script not up	−0.58 (−1.67 to +0.50)	yes
The check passes as often	12 of 12 (today's: 12)	yes
Controls clean	same: 0 broken; decoy: 0 coaching wins	yes
Verdict lines per script down	−4.5 (−5.1 to −4.0)	yes
and the asked moves made more often, points	+7 (−5 to +17)	**no**
At most 1.3 times as long	1.07 (0.97 to 1.17)	yes
Better moves as sound, points	0 (0 to 0)	yes

Table 1: The rule, fixed before the change's first run. Coaching and overall are the judge's scores per day; the claim rows, the asked moves and the unsound better moves are the audit's numbers per script, the change minus today's code; the verdict lines are the script's own count; the length is the change's script heard once over today's on the same day. Kept: no.

6 What the judge preferred

The judge preferred the change's coaching on all twelve days, and its whole script overall on all twelve: +1.00 (+1.00 to +1.00) on the train days and +1.00 (+1.00 to +1.00) held out. Truth went to the change on nine days and to today's code on zero, with three ties; listening to the change on ten

and to today's code on zero. Table 3 gives each day. Of the 92 decided verdicts in single orders, 53% went to the script shown second.

The controls held. Today's script against itself tied every criterion on every day (zero broken). The decoy, today's script with about 115 spoken words of restatement added, won coaching on zero days and lost listening on twelve and overall on twelve: the judge does not reward added words.

7 What the writing did

In the moment scenes	Today's code	The change
Says what the position is about before the move	53% (57)	70% (53)
Says what the better move does on the board	56% (57)	55% (53)
Gives an idea to reuse	68% (57)	77% (53)
Names a study lesson	14% (57)	43% (53)
and the lesson teaches what the scene teaches	75% (8)	74% (23)
The better move is sound by the evidence	100% (34)	100% (32)
Recites an engine line without saying what it does	7% (57)	9% (53)
Follows the better move to their best reply	68% (34)	66% (32)
Names the habit behind his move	49% (57)	51% (53)
Ends on the habit that replaces it	58% (57)	60% (53)
Any board scene: asks his move openly first	47% (96)	54% (97)
Praise names what the good move saw	87% (15)	92% (12)

Table 2: What the audit finds in the moment scenes, pooled over the twelve days: the share of the scenes that owe an item and make it, with the number owing it in brackets.

No verdict talk. Today's code spoke 4.5 lines a day that say who is better (three to six); the change spoke none. The audit, reading every board scene, found 4.9 scenes a day saying who is better in today's scripts and 0.3 in the change's.

The asked moves. The change's moment scenes said what the position is about before the move in 70% of them (today's: 53%) and gave an idea to reuse in 77% (68%), but said what the better move does on the board in 55% (56%). Per day, the share of the three it made rose by +7 (−5 to +17) points (Table 2). Every moment scene owes all three, including the scenes that show what he played before the better move, which is part of why neither config makes them in every scene.

Lessons. The change's annotations link a study lesson on 43% of their moments (37 of 87), today's on 13% (11 of 88). The audit found the change's moment scenes naming a lesson in 43% of them, and the lesson fitting what the scene teaches in 74% of those.

Truth. The change's scripts carried 0.42 false claims per script against 1.17 today, a difference of −0.75 (−1.42 to −0.08), and 1.08 false board claims against 1.67 (−0.58 (−1.67 to +0.50)). They carried more misleading claims, 3.17 per script against 2.25, and about as many unsupported ones (2.00 against 1.92). Most of the added misleading claims are of two kinds: a lesson's rule said in words the lesson does not use, and a line on how often his band finds a move followed by one that reads as

if he had played it. The better moves both taught were sound by the evidence in every moment scene that owed one.

Length, the check and cost. Heard once, the change's videos run 8.6 minutes on average against 8.2 (a ratio of 1.07 (0.97 to 1.17) on the same day). Every run passed the final check, in 2.3 check rounds on average against 2.3. The change ran `teach` 2.1 times a day (today's code: 1.7). A run cost 3.94 dollars against 3.32 and took 17 minutes against 14. The eval spent 123 dollars of a 180 budget (87 on writing, 36 on grading), and the subscription's seven-day window went from 13% to 19%.

			Judge, from the change's side				Length	Verdict lines		Asked moves	
Day	Split	Games	Truth	Coaching	Listening	Overall	ratio	Today	Change	Today	Change
Oct 13	train	14	+	+	+	+	0.80	5	0	100	61
Oct 14	held out	6	+	+	+	+	0.88	5	0	54	89
Oct 15	train	8	+	+	+	+	1.23	6	0	50	67
Oct 17	held out	5	+	+	+	+	0.94	3	0	67	75
Oct 18	train	1	0	+	+	+	1.03	3	0	67	50
Oct 19	held out	2	+	+	+	+	1.33	4	0	100	100
Oct 20	train	9	+	+	+	+	1.38	5	0	42	75
Oct 22	held out	9	0	+	+	+	1.05	4	0	53	73
Oct 24	train	1	+	+	+	+	1.08	5	0	42	50
Oct 25	held out	1	0	+	0	+	1.08	4	0	58	58
Oct 27	train	4	+	+	+	+	1.09	4	0	61	60
Oct 28	held out	2	+	+	0	+	0.92	6	0	50	67

Table 3: Each day: the judge's verdict on each criterion from the change's side (+ the change, − today's code, 0 a tie), the change's length over today's, the verdict lines each spoke, and the share of the asked moves each made.

8 Round 2: the scene of his move names the better move

Round 1's audit marks, for each moment scene, whether it teaches a better move and whether it says what that move does. Split that way (Table 4), the gap was structural: in both configs the scenes that teach a better move nearly always said what it does, and the scenes that show only his move never did, though the audit holds every moment scene to all three asked moves. Round 2 changes one thing, a sentence in the routine's prompt: a segment that plays his move in one scene and the better move in the next says what the better move is for in both, the scene of his move ending on the better move and what it does, in `teach`'s words, before the next scene plays it. The change is the app at `284a31b`, run as `24f056f` with the verdict check moved onto the replayed days as in round 1. Everything else was fixed before the run and is round 1's: the twelve days and their split, the writer, the graders and the rule. Today's code was not run again: round 2 reads round 1's baseline runs and their audits unchanged. The decoy was made again from today's script to the length of round 2's.

The rule. Every condition held (Table 5), so round 2 is kept. The judge preferred its coaching on eleven days and tied on one, and its whole script on eleven; truth went to round 2 on ten days, to today's code on one and tied on one (Table 7). The decoy, today's script with about 157 words of restatement added, won no day's coaching and lost listening on all twelve. Every run passed the final check.

Says what the better move does, in scenes that	teach a better move	show only his move
Today's code	94% (34)	0% (23)
The change, round 1	91% (32)	0% (21)
The change, round 2	100% (62)	–

Table 4: Moment scenes that say what the better move does, by whether the audit reads the scene as teaching a better move or as showing only his move, pooled over the twelve days, with the number of scenes in brackets. In round 2 every scene named a better move, so the audit read none as showing only his move.

Condition	Estimate (95% interval)	Holds
Coaching on the train days above 0	+0.83 (+0.50 to +1.00)	yes
Coaching on the held-out days above 0	+1.00 (+1.00 to +1.00)	yes
Overall not below 0	+0.92 (+0.75 to +1.00)	yes
False claims per script not up	+0.33 (−0.33 to +1.00)	yes
False board claims per script not up	+0.42 (−0.33 to +1.17)	yes
The check passes as often	12 of 12 (today's: 12)	yes
Controls clean	same: 0 broken; decoy: 0 coaching wins	yes
Verdict lines per script down	−4.5 (−5.1 to −4.0)	yes
and the asked moves made more often, points	+17 (+6 to +27)	yes
At most 1.3 times as long	1.11 (1.01 to 1.20)	yes
Better moves as sound, points	0 (0 to 0)	yes

Table 5: Round 2 against the rule fixed before round 1, over the twelve days unless a row says otherwise. The judge's rows are the mean verdict from round 2's side (+1 round 2, −1 today's code), the audit's rows per script, round 2 minus today's code.

The asked moves. Round 2's moment scenes said what the better move does in 100% of them (today's: 56%; round 1: 55%), said what the position is about first in 60% and gave an idea to reuse in 81% (Table 6). Per day the share of the three rose by +17 (+6 to +27) points, an interval clear of zero, where round 1's was +7 (−5 to +17).

What it cost. Round 2's scripts carried 1.50 false claims per script against 1.17 today (+0.33 (−0.33 to +1.00)) and round 1's 0.42, 2.08 false board claims against 1.67 (+0.42 (−0.33 to +1.17)), and 3.00 misleading ones against 2.25: more than today on each, inside the noise, where round 1 had fewer false claims than today. They followed the better move to their best reply in 34% of the scenes that owed it, against 68% today, but in about as many scenes (21 of 62, against 23 of 34): the scenes of his move now name a better move, so the audit holds them to following it too, and they leave that to the next scene. Its videos ran 8.9 minutes heard once, 1.11 (1.01 to 1.20) times today's on the same day, and spoke no verdict line. A run cost 3.54 dollars; the round spent 69 dollars and both rounds 192, under the 205 the owner approved for the second round.

In the moment scenes	Today's code	Round 1	Round 2
Says what the position is about before the move	53% (57)	70% (53)	60% (62)
Says what the better move does on the board	56% (57)	55% (53)	100% (62)
Gives an idea to reuse	68% (57)	77% (53)	81% (62)
Names a study lesson	14% (57)	43% (53)	37% (62)
and the lesson teaches what the scene teaches	75% (8)	74% (23)	74% (23)
The better move is sound by the evidence	100% (34)	100% (32)	100% (62)
Follows the better move to their best reply	68% (34)	66% (32)	34% (62)

Table 6: What the audit finds in the moment scenes in each round, pooled over the twelve days: the share of the scenes that owe an item and make it, with the number owing it in brackets.

		Judge, from round 2's side				Length	Verdict	Asked moves	
Day	Split	Truth	Coaching	Listening	Overall	ratio	lines	Today	Round 2
Oct 13	train	−	0	0	0	0.79	0	100	83
Oct 14	held out	+	+	+	+	1.15	0	54	86
Oct 15	train	+	+	+	+	1.29	0	50	83
Oct 17	held out	+	+	+	+	0.89	0	67	67
Oct 18	train	+	+	+	+	1.23	0	67	67
Oct 19	held out	+	+	+	+	1.32	0	100	91
Oct 20	train	+	+	+	+	1.28	0	42	83
Oct 22	held out	+	+	+	+	1.13	0	53	76
Oct 24	train	+	+	+	+	1.15	0	42	75
Oct 25	held out	0	+	+	+	0.97	0	58	67
Oct 27	train	+	+	+	+	1.14	0	61	100
Oct 28	held out	+	+	+	+	0.95	0	50	67

Table 7: Each day in round 2: the judge's verdict on each criterion from round 2's side (+ round 2, − today's code, 0 a tie), the length over today's, the verdict lines it spoke, and the share of the asked moves each made.

9 What the app can take

- By the rule, round 1 is not kept and round 2 is: with the sentence that ends the scene of his move on the better move and what it does, every condition held.
- The owner judged a day replayed with round 2 short of what he asked for: its scenes say what the better move is for, but a lesson is named rather than taught, and the line is played without saying why each move works, so it builds no pattern he can recognise. The audit's items do not ask for that, which is why the rule passed a version he did not want.
- The next version reads every move of the line his move allowed and of the line the pick starts on the board (a check, a trade, a take-back, a threat, what the move is for), and its prompt teaches the lesson's method on his position before any move, plays his move's line to what it costs, gives each move of the better line its job, and ends on the cue that the idea is on. Measuring it needs items for those, an audit that each move's stated job matches the board, and the owner's own judgement of a replayed day.
- False claims rose a little in round 2, inside the noise. A version that says more about each move has more to get wrong, so its round should hold false board claims at least as tightly as this rule does.

10 Caveats

- **One run per config and day.** There is no second run of today's code for a noise floor; the same-script and decoy controls stand in for it.
- **A model judges.** The judge and the audit are Sonnet 5.5 with the evidence, not the player; the controls show the judge neither invents differences nor rewards length.
- **The days are past days.** The writer reads the same data the routine read, but the lessons' server serves today's lessons.
- **Round 2 was designed after round 1.** Its sentence answers a gap round 1's audits of the same days showed, held-out days included, so round 2 is not an independent test: a new set of days would be.